

Mapping Meiotic Single-Strand DNA Reveals a New Landscape of DNA Double-Strand Breaks in *Saccharomyces cerevisiae*

SCIFACT-EVIDENCE-10874408

Evidence abstract

DNA double-strand breaks (DSBs), which are formed by the Spo11 protein, initiate meiotic recombination.

Previous DSB-mapping studies have used *rad50S* or *sae2* mutants, which are defective in break processing, to accumulate Spo11-linked DSBs, and report large (50 kb) "DSB-hot" regions that are separated by "DSB-cold" domains of similar size.

Substantial recombination occurs in some DSB-cold regions, suggesting that DSB patterns are not normal in *rad50S* or *sae2* mutants.

We therefore developed a novel method to map genome-wide, single-strand DNA (ssDNA)-associated DSBs that accumulate in processing-capable, repair-defective *dmc1* and *dmc1 rad51* mutants.

DSBs were observed at known hot spots, but also in most previously identified "DSB-cold" regions, including near centromeres and telomeres.

Although approximately 40% of the genome is DSB-cold in *rad50S* mutants, analysis of meiotic ssDNA from *dmc1* shows that most of these regions have substantial DSB activity.

Southern blot assays of DSBs in selected regions in *dmc1* , *rad50S*, and wild-type cells confirm these findings.

Thus, DSBs are distributed much more uniformly than was previously believed.

Comparisons of DSB signals in *dmc1*, *dmc1 rad51*, and *dmc1 spo11* mutant strains identify Dmc1 as a critical strand-exchange activity genome-wide, and confirm previous conclusions that Spo11-induced lesions initiate all meiotic recombination.

Document metadata

Corpus | SciFact

Document ID | 10874408

Evidence checklist

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Source continuity

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